AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM356
Interaction Design and User Experience

Mid-term Examination MTA - Makeup
Fall Semester 2025/2026

Date: 3/Dec. /2025

Number of Exam Pages: 3 (including this cover sheet)	Time Allowed: 2 hours

Instructions:

- Total Marks: **60**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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PART 1: Write True/False ANSWERS FOR ALL QUESTIONS [10 Marks]

1. Feedback in interaction design refers to providing users with information about the results of their actions and system status. (.....)
2. High-fidelity prototypes are most suitable for early conceptual design phases and initial user research activities. (.....)
3. Effective sketches help designers and users discuss key requirements and highlight differences between user needs. (.....)
4. Universal design principles ensure that products are accessible only to users with specific disabilities. (.....)
5. Heuristic evaluation requires actual users to test the interface and provide feedback on usability issues. (.....)
6. Iterative design involves cycles of design, evaluation, and refinement based on user feedback and testing results. (.....)
7. Think-aloud protocols during user testing provide insights into users' mental models and decision-making processes. (.....)
8. Good interaction design allows users to complete their goals efficiently, accurately, and effortlessly. (.....)
9. Reducing cognitive load always requires removing all visual elements and minimizing interface complexity. (.....)
10. The main goal of interaction design is to create products focused on aesthetic appeal and visual design. (.....)

PART 2: ANSWER ALL QUESTIONS [20 Marks]

Question 1: [10 marks]

You have been assigned to design a smart healthcare appointment booking system for a large hospital network.

Describe one way you could use direct and one way to use indirect observation to gather data to establish system requirements.

In each case, justify your answer in terms of the type of data that the technique would allow you to gather and what limitations it might have. [10 marks]

Question 2: [10 marks]

A) Why is cognitive load considered important in interaction design? Illustrate your answer with an example. [5 marks]

B) What are the Codes of Professional Conduct for Usability Practitioners? Highlight five. [5 marks]

PART 3: ANSWER ALL QUESTIONS [30 marks]

Question 1: [15 marks]

You are designing a smart public transportation app that allows commuters to plan routes, check real-time bus/train schedules, purchase tickets, and receive service alerts.

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A) Identify two methods you could use to gather data for creating accurate user personas for this system. Briefly explain how each method contributes to understanding commuter needs. [5 marks]
B) Create one example persona for a typical commuter using the smart transportation app. Explain how this persona can guide specific design decisions to improve usability, accessibility, and engagement. [10 marks]

Question 2: [15 marks]

You are a UX Designer evaluating "FitTrack Pro," a fitness and wellness mobile app designed to help users track workouts, monitor nutrition, set fitness goals, and connect with personal trainers.
Recently, users have reported issues with confusing workout logging processes, delayed synchronization with wearable devices, and difficulty interpreting nutrition data visualizations. You have been asked to analyze the user experience challenges and propose improvement strategies.

A) Identify and explain three significant user experience challenges that users might encounter while using "FitTrack Pro." [4.5 marks]
B) How might these challenges impact user motivation and long-term engagement with fitness goals? [1.5 marks]
C) Critically evaluate how the current design approach of "FitTrack Pro" aligns with principles of interaction design and usability. [2 marks]
D) What effect could the current design have on novice users or those with limited fitness knowledge? [2 marks]
E) Propose three strategic UX improvements that can enhance workout logging, device synchronization, and data visualization in "FitTrack Pro." [3 marks]
F) Describe how these improvements could address the identified issues and improve user retention and fitness outcomes. [2 marks]

End of questions

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